

Neoadjuvant Chemotherapy with Trastuzumab plus Pertuzumab versus Trastuzumab Alone in Early HER2-Positive Breast Cancer: A Single-Center Retrospective Cohort Study

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This demonstration manuscript was written by AI (Claude Opus 5.5). In a real submission, the text must be written by the authors.

Abstract

Background: Randomized trials showed that adding pertuzumab to neoadjuvant trastuzumab and chemotherapy increases pathologic complete response (pCR) in human epidermal growth factor receptor 2 (HER2)-positive breast cancer, but real-world comparisons across treatment eras are limited. We compared dual and single HER2 blockade in routine practice.

Materials and Methods: This single-center retrospective cohort study emulated a target trial in patients with clinical stage II–III HER2-positive breast cancer who started neoadjuvant chemotherapy with trastuzumab between January 2012 and December 2023. The primary outcome was pCR (ypT0/is ypN0); isolated tumor cells in lymph nodes were not considered pCR. Because the two regimens were used in largely different years, the groups were balanced with propensity-score overlap weighting. Estimates are reported with 95% confidence intervals (CIs).

Results: Of 615 patients, 416 received trastuzumab plus pertuzumab and 199 received trastuzumab alone. Unweighted pCR rates were 195/416 (46.9%) and 90/199 (45.2%), respectively. After weighting, pCR rates were 57.6% and 40.3% (odds ratio, 2.01 [95% CI, 1.12–3.62]; $P = 0.019$). With a median follow-up of 6.2 years, the hazard ratios were 0.89 (95% CI, 0.47–1.69) for event-free survival and 0.76 (95% CI, 0.34–1.66) for overall survival.

Conclusion: Dual HER2 blockade was associated with a higher pCR rate, consistent with randomized trials, but the estimate depended on adjustment for treatment era. The data were insufficient to assess event-free or overall survival.

Keywords: Breast neoplasms, Receptor, ErbB-2, Neoadjuvant therapy, Pertuzumab, Trastuzumab, Propensity score

Introduction

Adding pertuzumab to neoadjuvant trastuzumab and chemotherapy increased the rate of pathologic complete response (pCR) in randomized trials of human epidermal growth factor receptor 2 (HER2)-positive breast cancer.^[1,2] Patients who achieve pCR have longer event-free survival,^[3] but the long-term benefit of pertuzumab is modest.^[4] Observational studies have reported higher pCR rates with dual HER2 blockade,^[5] yet the two regimens were often used in different years, which complicates comparison. We compared neoadjuvant trastuzumab plus pertuzumab with trastuzumab alone in routine practice, using methods designed for limited overlap between treatment groups.

Materials and Methods

Study Design and Data Sources

This single-center retrospective cohort study was designed to emulate a target trial comparing two neoadjuvant treatment strategies.^[6] Data were extracted from the hospital cancer registry, electronic drug orders, surgical pathology reports, and follow-up records. Before analysis, the extracts were checked for completeness, plausible values, duplicates, and the order of dates; errors identified during these checks were queried with the hospital information office and corrected from source documents. Medical record numbers were replaced with random study identifiers before analysis. The study is reported according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement.^[7]

Patients

We included patients aged 20 years or older with clinical stage II–III breast cancer (American Joint Committee on Cancer, 8th edition) that was HER2-positive, defined as an

immunohistochemistry (IHC) score of 3+ or amplification on in situ hybridization (ISH), who started neoadjuvant chemotherapy with trastuzumab between January 2012 and December 2023. We excluded patients with distant metastasis at diagnosis, surgery before systemic therapy, neoadjuvant trastuzumab emtansine (T-DM1), or surgery at another hospital without an available pathology report. Follow-up data were available until December 31, 2025.

Treatment Groups and Outcomes

The index date was the date of the first neoadjuvant anticancer drug order. Patients were assigned to the dual-blockade group if any pertuzumab order, including the fixed-dose subcutaneous combination of pertuzumab and trastuzumab, occurred between the index date and the day before surgery, and to the single-blockade group otherwise.

The primary outcome was pCR, defined as no residual invasive cancer in the breast and axillary lymph nodes (ypT0/is ypN0). In line with international recommendations, isolated tumor cells (ypN0(i+)) and micrometastases in lymph nodes were not considered pCR.^[8,9] Patients whose disease progressed before surgery were classified as not achieving pCR. The secondary outcome was event-free survival (EFS), measured from the index date to disease progression precluding surgery, locoregional recurrence, distant metastasis, or death from any cause. Overall survival (OS) was an exploratory outcome. Patients without an event were censored at the later of the last clinic visit and the last drug order, or at the follow-up cutoff, whichever came first.

Statistical Analysis

The statistical analysis plan was finalized before any outcome was examined, and its differences from the original protocol were approved by the institutional review board as a protocol amendment. The propensity score for receiving dual blockade was estimated by logistic regression with 13 baseline covariates: age, menopausal status, Eastern Cooperative Oncology

Group performance status, body mass index, baseline left ventricular ejection fraction, clinical T and N categories, hormone receptor status, HER2 IHC score, histologic grade, Ki-67 index, anthracycline-based chemotherapy, and year of treatment start. Because the two regimens were used in largely different periods, we used overlap weighting, which emphasizes patients for whom either treatment was plausible and avoids extreme weights.^[10–12] Covariate balance was assessed with standardized mean differences (SMDs), with values below 0.1 indicating adequate balance. Missing covariate values were imputed by multiple imputation with 20 data sets that excluded outcome variables, and estimates were combined with Rubin’s rules.^[13] The pCR rates were compared with weighted logistic regression, reported as odds ratios (ORs), and with weighted risk differences; standard errors accounted for estimation of the propensity score. EFS and OS were estimated with weighted Kaplan-Meier curves and compared with weighted Cox models, reported as hazard ratios (HRs) with robust standard errors; the proportional hazards assumption was assessed with Schoenfeld residuals. Prespecified sensitivity analyses restricted the cohort to patients who started treatment in 2016–2019, used one-to-one propensity-score matching, used multivariable regression, defined pCR as ypT0 ypN0, and included only patients with complete covariate data. An exploratory analysis tested whether the association with pCR differed by hormone receptor status. The E-value was calculated to quantify how strong unmeasured confounding would need to be to explain away the primary result.^[14] Two-sided *P* values below 0.05 were considered statistically significant. Analyses were performed with R version 4.5.1 using the WeightIt, MatchIt, cobalt, mice, survival, and EValue packages.

Ethics

The study was approved by the institutional review board of the participating hospital, which waived the requirement for informed consent because of the retrospective design and the use of de-identified data. The study was conducted in accordance with the Declaration of Helsinki (1975, revised in 2013).

Results

Patients

Of 703 registry records, three duplicate records were removed, leaving 700 patients (Figure 1). After exclusion of patients with clinical stage 0–I disease ($n = 27$), distant metastasis at diagnosis ($n = 14$), HER2-negative disease ($n = 11$), no systemic therapy at the study hospital ($n = 17$), and surgery at another hospital ($n = 16$), 615 patients were included: 416 received trastuzumab plus pertuzumab and 199 received trastuzumab alone.

Baseline characteristics were similar between the groups except for the year of treatment start (SMD, 2.80) and the use of anthracycline-based chemotherapy (SMD, 0.55) (Table 1). Most patients who received trastuzumab alone started treatment in 2012–2015, whereas most patients who received dual blockade started treatment in 2020–2023. After overlap weighting, all covariates were exactly balanced (maximum absolute SMD, 0.000), and the effective sample sizes were 86 and 66 patients, respectively. The median follow-up was 6.2 years overall, 5.1 years for dual blockade, and 10.7 years for trastuzumab alone.

Pathologic Complete Response

In the unweighted data, pCR was achieved in 195/416 (46.9%) patients who received dual blockade and in 90/199 (45.2%) patients who received trastuzumab alone (Table 2). Within

treatment periods, however, crude pCR rates were higher with dual blockade (2012–2015: 3/6 (50.0%) vs. 73/156 (46.8%); 2016–2019: 90/171 (52.6%) vs. 17/38 (44.7%)), and the overall pCR rate in 2020–2023, when only five patients received trastuzumab alone, was 102/244 (41.8%). After overlap weighting, the pCR rate was 57.6% with dual blockade and 40.3% with trastuzumab alone, a difference of 17.3 percentage points (95% CI, 3.1–31.5); the OR was 2.01 (95% CI, 1.12–3.62) ($P = 0.019$).

Event-Free and Overall Survival

EFS events occurred in 62 patients who received dual blockade and in 68 patients who received trastuzumab alone. The weighted five-year EFS rates were 88.6% and 81.2%, respectively (HR, 0.89 [95% CI, 0.47–1.69]; $P = 0.718$) (Table 3, Figure 2). Deaths occurred in 34 and 53 patients, and the weighted five-year OS rates were 89.6% and 86.9% (HR, 0.76 [95% CI, 0.34–1.66]; $P = 0.488$) (Figure 3). There was no evidence against proportional hazards for EFS ($P = 0.394$) or OS ($P = 0.462$). Among patients who did not achieve pCR, adjuvant T-DM1 was given to 59.7% in the dual-blockade group and 4.6% in the single-blockade group.

Sensitivity and Subgroup Analyses

The direction of the association with pCR was consistent across sensitivity analyses (Table 2): the OR was 1.97 (95% CI, 1.10–3.53) with multivariable regression and 2.38 (95% CI, 1.27–4.46) when pCR was defined as ypT0 ypN0. In the cohort restricted to 2016–2019 (171 and 38 patients) and in the complete-case analysis, the ORs were 2.15 (95% CI, 0.92–5.01) and 1.91 (95% CI, 0.99–3.68), respectively, with confidence intervals that included the null value.

Propensity-score matching retained 44 pairs and yielded a smaller estimate (OR, 1.33 [95% CI, 0.52–3.43]). The association did not differ by hormone receptor status (receptor-negative OR, 2.63 [95% CI, 0.79–8.71]; receptor-positive OR, 2.03 [95% CI, 0.94–4.37]; $P = 0.552$ for

interaction). The E-value for the primary OR was 2.19, and that for its lower confidence limit was 1.31.

Discussion

In this single-center cohort, neoadjuvant trastuzumab plus pertuzumab was associated with a higher pCR rate than trastuzumab alone after measured characteristics were balanced with overlap weighting. The size of the difference was similar to that observed in randomized trials, but the evidence for EFS and OS was inconclusive.

In NeoSphere, adding pertuzumab to trastuzumab and docetaxel increased the breast pCR rate from 29.0% to 45.8%,^[1] and the Asian PEONY trial reported a similar increase in total pCR,^[2] with five-year EFS of 84.8% versus 73.7% in its descriptive final analysis.^[15] Real-world studies from Turkey, Italy, China, and Spain have also reported higher pCR rates with dual blockade,^[16–19] and a meta-analysis of 18 real-world studies estimated a pooled OR of 1.81 (95% CI, 1.56–2.09).^[5] Our estimate is consistent with these findings.

Our results also show how strongly treatment era can influence real-world comparisons. The two regimens were used in largely different years, and the crude pCR rates were almost identical because the overall pCR rate was lower in recent years, when nearly all patients received dual blockade. Within each period, however, pCR was more frequent with dual blockade. We could not determine why the pCR rate was lower in recent years; possible explanations include changes in patient selection, chemotherapy backbones, and pathologic assessment, such as more thorough examination of lymph nodes that identifies isolated tumor cells as residual disease. Overlap weighting addresses this problem by focusing the comparison on patients who were treated when either regimen was plausible, mainly those treated in 2016–2019, and avoids the extreme weights

that inverse probability weighting would have produced.^[11] The estimate should therefore be interpreted as the effect in patients for whom both regimens were realistic options.

The EFS and OS results were inconclusive. Follow-up was shorter in the dual-blockade group, events were few, and post-neoadjuvant treatment changed over time. After the KATHERINE trial showed that adjuvant T-DM1 improves invasive disease-free survival and OS in patients with residual disease,^[20,21] T-DM1 was used mainly in the dual-blockade era in our cohort. In APHINITY, adjuvant pertuzumab did not improve OS at a median follow-up of 8.4 years,^[4] and pCR has not been validated as a trial-level surrogate for EFS or OS.^[3] A higher pCR rate should therefore not be assumed to translate into a proportional survival benefit.

We applied the international definition of pCR, under which residual isolated tumor cells are considered residual disease.^[8,9] Low-volume residual nodal disease after neoadjuvant chemotherapy, including isolated tumor cells, has been associated with worse disease-free and overall survival than node-negative disease,^[22] which supports this stricter definition.

The strengths of this study include an analysis plan that was finalized before outcomes were examined, validation and correction of the source data before analysis, a design that addressed the limited overlap between treatment eras, and several sensitivity analyses. This study also has limitations. First, it was a single-center retrospective study, and residual confounding by unmeasured factors associated with treatment era cannot be excluded; an unmeasured confounder associated with both treatment and pCR by an OR of approximately 2.19 could explain the observed association. Second, the effective sample size after weighting was small, so the estimates were imprecise, and the matched analysis, which retained only 44 pairs, yielded a smaller estimate. Third, EFS did not include contralateral breast cancer or second primary cancers because these events were not recorded, and loss to follow-up may have differed between treatment eras. Fourth, data on other active malignancies and on participation in clinical trials

were unavailable. Finally, missing covariate data were handled by multiple imputation, which assumes that the data were missing at random.

In conclusion, in routine practice, adding pertuzumab to neoadjuvant trastuzumab and chemotherapy was associated with a higher pCR rate, consistent with randomized evidence. The result depended on adjustment for treatment era, and the data were insufficient to determine the effects on EFS or OS.

Data Availability Statement

The datasets generated and/or analyzed during the current study are not publicly available due to patient privacy regulations but are available from the corresponding author on reasonable request. In this teaching demonstration, all data are synthetic.

Use of Generative AI

This demonstration manuscript, including all of its text, was written by an AI model (Claude Opus 5.5, Anthropic) at the request of the authors for teaching purposes; in a real submission, the text must be written by the authors, as required by the journal's policy on generative AI. Claude Opus 5.5 was also used, under the authors' direction, to write and test the statistical analysis code, check the data, produce the tables and figures, and format the manuscript. OpenEvidence was used to search the literature, and every reference was verified against Crossref and PubMed. The analysis plan was specified by the authors before outcomes were examined and was approved by the institutional review board. The authors are responsible for reviewing all code, results, and text and take full responsibility for the content. The analysis code can be provided as supplementary material.

References

1. Gianni L, Pienkowski T, Im YH, Roman L, Tseng LM, Liu MC et al. Efficacy and safety of neoadjuvant pertuzumab and trastuzumab in women with locally advanced, inflammatory, or early HER2-positive breast cancer (NeoSphere): a randomised multicentre, open-label, phase 2 trial. *Lancet Oncol* 2012;13:25-32.
2. Shao Z, Pang D, Yang H, Li W, Wang S, Cui S et al. Efficacy, Safety, and Tolerability of Pertuzumab, Trastuzumab, and Docetaxel for Patients With Early or Locally Advanced ERBB2-Positive Breast Cancer in Asia: The PEONY Phase 3 Randomized Clinical Trial. *JAMA Oncol* 2020;6:e193692.
3. Cortazar P, Zhang L, Untch M, Mehta K, Costantino JP, Wolmark N et al. Pathological complete response and long-term clinical benefit in breast cancer: the CTNeoBC pooled analysis. *Lancet* 2014;384:164-72.
4. Loibl S, Jassem J, Sonnenblick A, Parlier D, Winer E, Bergh J et al. Adjuvant Pertuzumab and Trastuzumab in Early Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer in the APHINITY Trial: Third Interim Overall Survival Analysis With Efficacy Update. *J Clin Oncol* 2024;42:3643-51.
5. Matsas S, Zembala J, Ruiz Simões A, Giuliani Schmitt L, Abdou Y, Del Giglio A. Pertuzumab and Pathological Complete Response in Early HER2+ Breast Cancer: A Systematic Review and Meta-analysis of Real-World Studies. *Asian Pac J Cancer Prev* 2026;27:813-21.
6. Hernán MA, Robins JM. Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available. *Am J Epidemiol* 2016;183:758-64.
7. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet* 2007;370:1453-7.

8. Provenzano E, Bossuyt V, Viale G, Cameron D, Badve S, Denkert C et al. Standardization of pathologic evaluation and reporting of postneoadjuvant specimens in clinical trials of breast cancer: recommendations from an international working group. *Mod Pathol* 2015;28:1185-201.
9. Bossuyt V, Provenzano E, Symmans WF, Webster F, Allison KH, Dang C et al. A dedicated structured data set for reporting of invasive carcinoma of the breast in the setting of neoadjuvant therapy: recommendations from the International Collaboration on Cancer Reporting (ICCR). *Histopathology* 2024;84:1111-29.
10. Li F, Morgan KL, Zaslavsky AM. Balancing Covariates via Propensity Score Weighting. *Journal of the American Statistical Association* 2018;113:390-400.
11. Li F, Thomas LE, Li F. Addressing Extreme Propensity Scores via the Overlap Weights. *Am J Epidemiol* 2019;188:250-7.
12. Thomas LE, Li F, Pencina MJ. Overlap Weighting: A Propensity Score Method That Mimics Attributes of a Randomized Clinical Trial. *JAMA* 2020;323:2417-8.
13. Leyrat C, Seaman SR, White IR, Douglas I, Smeeth L, Kim J et al. Propensity score analysis with partially observed covariates: How should multiple imputation be used? *Stat Methods Med Res* 2019;28:3-19.
14. VanderWeele TJ, Ding P. Sensitivity Analysis in Observational Research: Introducing the E-Value. *Ann Intern Med* 2017;167:268-74.
15. Huang L, Pang D, Yang H, Li W, Wang S, Cui S et al. Neoadjuvant-adjuvant pertuzumab in HER2-positive early breast cancer: final analysis of the randomized phase III PEONY trial. *Nat Commun* 2024;15:2153.

16. Bilici A, Olmez OF, Kaplan MA, Oksuzoglu B, Sezer A, Karadurmus N et al. Impact of adding pertuzumab to trastuzumab plus chemotherapy in neoadjuvant treatment of HER2 positive breast cancer patients: a multicenter real-life HER2PATH study. *Acta Oncol* 2023;62:381-90.
17. Fabbri A, Nelli F, Botticelli A, Giannarelli D, Marrucci E, Fiore C et al. Pathologic response and survival after neoadjuvant chemotherapy with or without pertuzumab in patients with HER2-positive breast cancer: the Neopearl nationwide collaborative study. *Front Oncol* 2023;13:1177681.
18. Cheng Y, Xiang H, Xin L, Duan X, Liu Y. Neoadjuvant therapy for early human epidermal growth factor receptor 2 positive breast cancer in China: A multicenter real-world study (CSBrS-015). *Chin Med J (Engl)* 2022;135:2311-8.
19. Díaz-Redondo T, Lavado-Valenzuela R, Jimenez B, Pascual T, Gálvez F, Falcón A et al. Different Pathological Complete Response Rates According to PAM50 Subtype in HER2+ Breast Cancer Patients Treated With Neoadjuvant Pertuzumab/Trastuzumab vs. Trastuzumab Plus Standard Chemotherapy: An Analysis of Real-World Data. *Front Oncol* 2019;9:1178.
20. von Minckwitz G, Huang CS, Mano MS, Loibl S, Mamounas EP, Untch M et al. Trastuzumab Emtansine for Residual Invasive HER2-Positive Breast Cancer. *N Engl J Med* 2019;380:617-28.
21. Geyer CE Jr, Untch M, Huang CS, Mano MS, Mamounas EP, Wolmark N et al. Survival with Trastuzumab Emtansine in Residual HER2-Positive Breast Cancer. *N Engl J Med* 2025;392:249-57.
22. Wong SM, Almana N, Choi J, Hu J, Gagnon H, Natsuhara K et al. Prognostic Significance of Residual Axillary Nodal Micrometastases and Isolated Tumor Cells After Neoadjuvant Chemotherapy for Breast Cancer. *Ann Surg Oncol* 2019;26:3502-9.

Table 1. Baseline characteristics of the study cohort, by neoadjuvant anti-HER2 regimen

Characteristic	Trastuzumab + pertuzumab (n = 416)	Trastuzumab alone (n = 199)	SMD†
Age at diagnosis, years, median (IQR)	50.7 (43.8–59.0)	51.5 (42.6–57.7)	0.05
Postmenopausal	232 (55.8)	116 (58.3)	0.05
ECOG performance status ≥ 1	107 (25.7)	58 (29.1)	0.08
BMI, kg/m ² , median (IQR)	24.1 (21.5–26.8)	24.0 (21.2–26.4)	0.08
Baseline LVEF, %, median (IQR)	65.0 (61.0–68.0)	65.0 (62.0–68.0)	0.08
Clinical T category			0.09
T1	32 (7.7)	17 (8.5)	
T2	237 (57.0)	111 (55.8)	
T3	97 (23.3)	42 (21.1)	
T4	50 (12.0)	29 (14.6)	
Clinical N category			0.00
N0	122 (29.3)	58 (29.1)	
N1	196 (47.1)	94 (47.2)	
N2	56 (13.5)	27 (13.6)	
N3	42 (10.1)	20 (10.1)	
Hormone receptor positive (ER or PR $\geq 1\%$)	240 (57.7)	116 (58.3)	0.01
HER2 IHC 3+ (vs IHC 2+, ISH amplified)	320 (76.9)	155 (77.9)	0.02
Histologic grade 3*	201 (51.9)	88 (48.6)	0.07
Ki-67, %, median (IQR)*	43.5 (31.2–56.0)	43.5 (30.0–56.8)	0.05
Anthracycline-based chemotherapy	116 (27.9)	107 (53.8)	0.55
Year of neoadjuvant therapy start			2.80
2012–2015	6 (1.4)	156 (78.4)	
2016–2019	171 (41.1)	38 (19.1)	
2020–2023	239 (57.5)	5 (2.5)	

Values are n (%) unless otherwise indicated. Abbreviations: BMI, body mass index; ECOG, Eastern Cooperative Oncology Group; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; IQR, interquartile range; ISH, in situ

hybridization; LVEF, left ventricular ejection fraction; PR, progesterone receptor; SMD, standardized mean difference.

*Missing: histologic grade, 47; Ki-67, 71. Percentages exclude missing values.

†Before weighting. After overlap weighting, every covariate in the propensity-score model was exactly balanced (maximum absolute SMD, 0.000).

Table 2. Pathologic complete response: primary and sensitivity analyses

Analysis	Patients, n (T+P/T)	pCR, T+P	pCR, T	Risk difference, percentage points (95% CI)	Odds ratio (95% CI)	P
Unweighted	416/199	195 (46.9)	90 (45.2)			
Overlap weighting (primary)	416/199	57.6%*	40.3%*	17.3 (3.1–31.5)	2.01 (1.12–3.62)	0.019
S1: Treatment start 2016–2019 only	171/38	63.1%*	44.3%*	18.8 (–1.7 to 39.3)	2.15 (0.92–5.01)	0.076
S2: 1:1 propensity-score matching	44/44	53.9%*	46.8%*	7.1 (–16.4 to 30.6)	1.33 (0.52–3.43)	0.552
S3: Multivariable regression	416/199				1.97 (1.10–3.53)	0.024
S4: pCR defined as ypT0 ypN0	416/199	40.4%*	22.1%*	18.2 (5.6–30.9)	2.38 (1.27–4.46)	0.007
S6: Complete cases	333/164	56.2%*	40.3%*	16.0 (0.0–32.0)	1.91 (0.99–3.68)	0.054

Unweighted row: n (%) of patients with pCR. Abbreviations: CI, confidence interval; pCR, pathologic complete response (ypT0/is ypN0; isolated tumor cells in lymph nodes were not considered pCR); T, trastuzumab; T+P, trastuzumab plus pertuzumab.

*Weighted (overlap weighting) or matched proportions, averaged over 20 imputations; no absolute counts correspond to weighted proportions.

S3 adjusts for the propensity-score covariates by regression and has no corresponding weighted proportions or risk difference.

Table 3. Event-free and overall survival

Analysis	Events, n (T+P/T)	Hazard ratio (95% CI)	<i>P</i>	5-year rate, T+P	5-year rate, T
Event-free survival†					
Overlap weighting (primary)	62/68	0.89 (0.47–1.69)	0.718	88.6%*	81.2%*
S1: Treatment start 2016–2019 only	37/9	0.74 (0.29–1.90)	0.536	89.7%*	75.7%*
S2: 1:1 propensity-score matching	9.3‡/11.2‡	0.85 (0.34–2.13)	0.725		
S3: Multivariable regression	62/68	0.94 (0.53–1.70)	0.847		
S6: Complete cases	47/56	1.00 (0.48–2.07)	0.998	91.0%*	85.1%*
Overall survival					
Overlap weighting (primary)	34/53	0.76 (0.34–1.66)	0.488	89.6%*	86.9%*
S1: Treatment start 2016–2019 only	21/8	0.56 (0.19–1.61)	0.282	90.1%*	82.7%*
S2: 1:1 propensity-score matching	7.3‡/10.8‡	0.75 (0.28–2.04)	0.571		
S3: Multivariable regression	34/53	0.68 (0.32–1.45)	0.321		
S6: Complete cases	26/43	0.94 (0.38–2.33)	0.889	91.6%*	87.6%*

Abbreviations: CI, confidence interval; T, trastuzumab; T+P, trastuzumab plus pertuzumab.

*Weighted Kaplan-Meier estimates, averaged over 20 imputations.

†Events: progression precluding surgery, locoregional recurrence, distant metastasis, or death from any cause.

‡Matched analysis: mean number of events across imputations.

Figure Legends

Figure 1. Flow diagram of patient selection. HER2, human epidermal growth factor receptor 2.

Figure 2. Overlap-weighted Kaplan-Meier curves of event-free survival by neoadjuvant regimen, averaged over 20 imputations, with the numbers of patients at risk. CI, confidence interval; HR, hazard ratio.

Figure 3. Overlap-weighted Kaplan-Meier curves of overall survival by neoadjuvant regimen, averaged over 20 imputations, with the numbers of patients at risk. CI, confidence interval; HR, hazard ratio.